

ANIMARKET

A MOBILE LIVESTOCK MARKETPLACE FOR DAVAO DEL NORTE

Final Project Documentation
Version 1.0.0



Department of Computing Education
UM Tagum College, Tagum City

CCE106 - APPLICATION DEVELOPMENT & EMERGING TECHNOLOGY

**DWAYNE RUBITE
CHESTER EMUTAN**

October 2026
Current app snapshot as of 7 October 2026

Table of Contents

Chapter 1 Introduction and project overview	3
Requirements	4
Chapter 2 Architecture and technology stack	5
Design and user interface	6
Database design	7
API design and development process	8
Chapter 3 Testing	9
Deployment and user acceptance	10
Maintenance and conclusion	11
References	12

List of Tables

Table 1 Functional requirements	4
Table 2 Technology stack	5
Table 3 Primary data collections	7
Table 4 Trusted API interfaces	8
Table 5 Key test cases and evidence	9
Table 6 Maintenance plan	11

List of Figures

Figure 1 AniMarket system architecture	5
Figure 2 Order lifecycle	6

CHAPTER 1

Introduction

Purpose and audience

AniMarket is a mobile livestock marketplace for Davao del Norte. It brings livestock discovery, seller listing management, price guidance, private communication and order coordination into one account. This documentation explains the application for course evaluators, project developers, future maintainers and marketplace stakeholders.

Project overview

Project: AniMarket

Developers: Dwayne Rubite and Chester Emutan

Course: CCE106 - Application Development & Emerging Technology

Institution: Department of Computing Education, UM Tagum College, Tagum City

App version: 1.0.0

Documentation date: 7 October 2026

The current application targets Android, iOS and web using Expo SDK 57. A customer account can act as both buyer and seller. Identity approval is required before publishing livestock or placing an order. Authorized staff use a separate administration flow for ID reviews and, with the required permissions, market-price maintenance.

Implemented scope

The marketplace covers Cow, Pig, Goat and Chicken listings. Customers can search and filter livestock, publish and manage their own listings, submit vaccination evidence, view dated market references, calculate advisory prices, exchange messages, coordinate orders and view in-app notifications. Public user profiles and editable star ratings are also available to signed-in customers.

Configured cloud builds use Firebase Authentication and Firestore, a trusted Supabase Edge Function, private Supabase document storage and Cloudinary public photos. Default sample catalogs have been removed; an empty feed is valid until sellers publish records.

Current operating limits

The supported trading area is Davao del Norte, including Samal. Payment and transport are buyer-seller arrangements; the app does not process payments or book couriers. Vaccination evidence is seller-provided. Native voice audio and final device acceptance still need physical-device testing. Full offline operation, background incoming-call push and email ownership verification are not implemented.

Requirements

Table 1 Functional requirements

ID	Requirement	Current behavior
FR01	Accounts	Register, sign in, retain sessions and safely sign out.
FR02	Unified account	Buy and sell with the same customer account.
FR03	Identity review	Complete the address and obtain ID approval to publish or order.
FR04	Discovery	Browse/search listings and filter or sort matching livestock.
FR05	Listings	Use up to four photos; manage owned drafts and listing states.
FR06	Vaccination	Record the date, vaccine/disease and private proof file.
FR07	Location	Confirm addresses and map pins within Davao del Norte.
FR08	Price guidance	Use dated sources and animal factors for advisory estimates.
FR09	Communication	Exchange private live text and supported voice calls.
FR10	Orders	Review fulfillment, payment arrangement and estimated totals.
FR11	Order status	Track seller actions, buyer completion and eligible cancellation.
FR12	Notifications	Show in-app activity and retain user-specific read state.
FR13	User profiles	View customer profiles and save one editable rating per user.
FR14	Profile settings	Edit personal/photo details and reauthenticate security changes.
FR15	Administration	Authorize ID review and source-checked price editing.

Non Functional Requirements

Security and privacy: Enforce authenticated ownership, participant access, private-document controls and server-authorized changes. **Reliability and integrity:** Handle retries, stale records, duplicate submissions, account switching and valid state transitions. **Usability and accessibility:** Provide readable labels, keyboard-aware forms, clear status text and predictable navigation. **Performance and maintainability:** Bound image uploads and message queries, paginate message history, and separate feature layers. **Compatibility and transparency:** Support platform-specific behavior and state the limits of references, ratings, health claims, delivery and voice.

CHAPTER 2

Architecture

Expo Router files in **src/app** adapt navigation. Features separate presentation, application, domain and data layers. Shared services and repositories connect screens to cloud data. The legacy Node/SQLite server is a development fallback.

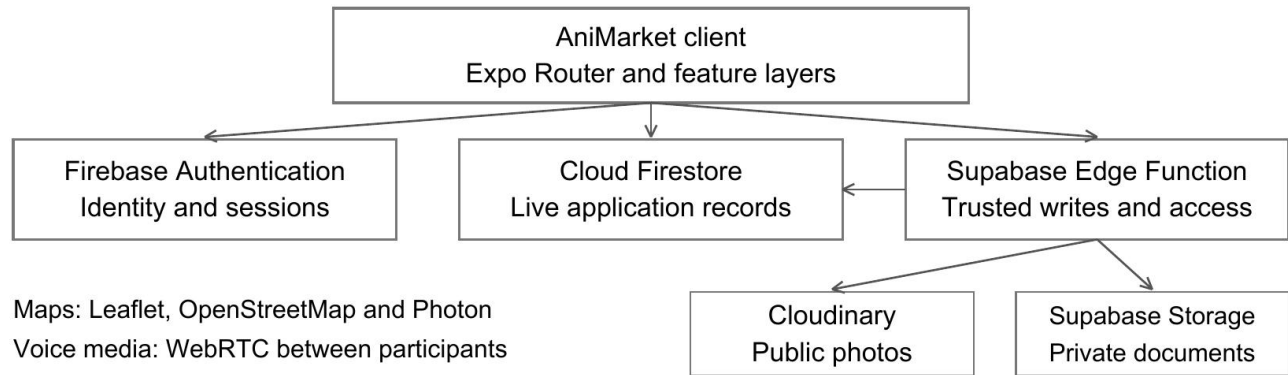


Figure 1 AniMarket system architecture

Table 2 Technology stack

Layer	Technology	Purpose
Application	Expo 57.0.26; React Native 0.86.3; React 19.2.3	Shared Android, iOS and web interface.
Navigation and language	Expo Router 57.0.24; TypeScript 6.0.3	File-based routes and typed feature code.
Identity and live data	Firebase 12.19.0; Firebase Auth; Firestore	Accounts, persistent records and live listeners.
Trusted backend	Supabase Edge Function	Token validation, authorization and business mutations.
Media	Cloudinary; private Supabase Storage	Public photos and restricted ID/vaccination documents.
Maps and voice	Leaflet; OpenStreetMap; Photon; WebRTC	Location selection and participant voice media.
Delivery and checks	EAS; Expo lint; TypeScript; Node checks	Configured builds and automated validation.

High level data flow

Firebase authenticates the client. Firestore listeners deliver public or participant-authorized reads. For trusted writes, the client sends a Firebase token to Supabase, which checks identity/permissions, applies business rules and saves records. Firestore then delivers updates. Public-photo uploads and private-document operations use separate media paths.

Design

User interface design

Portrait layouts use a light theme, consistent navigation and keyboard-aware forms. Loading, empty, failure and retry states are explicit. Direct links recover when a record is missing.

Account entry and identity

Registration collects first and last names, optional middle name, phone, email, matching passwords and terms consent. Passwords require 8-21 characters with uppercase, lowercase, number and symbol. Address completion occurs in Personal Information. Account Verification accepts a valid photo ID with consent and shows pending, approved, rejected or expired status.

Marketplace and selling

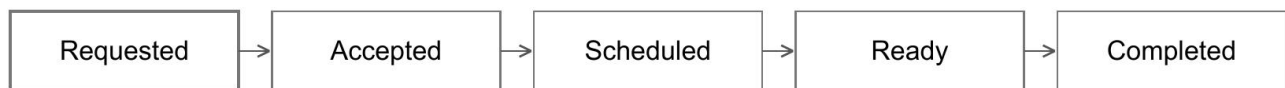
Home shows active listings; Search and Filter narrows results and changes sort order. Listing Details presents animal information, asking price, locality, seller context and contact/order actions. My Listings manages owned records. Publication requires a complete address, approved identity, photos and a confirmed pickup pin. Reservation prevents conflicting listing changes.

Pricing and profiles

Market Reference shows dated averages/ranges, units, source links and history. The calculator uses live weight, reference rates and condition/age/purpose factors; its sex selector does not affect the formula. Customer profiles show address/contact fields and editable ratings, which are not verified-purchase reviews.

Orders and communication

Checkout collects a future pickup date/time or delivery date/address/receiver details. Review requires agreement before submission. My Orders and Order Status display persisted participant records and allowed actions. Messages provide private text history and supported voice controls; notifications open the related activity.



Delivery includes In transit between Ready and Completed.

Decline or eligible cancellation ends the request without completing a sale.

Figure 2 Order lifecycle

Administration

Staff login opens Admin Portal. ID Reviews supports private preview, approval or rejection with feedback. Market Prices requires both staff and reviewer permissions, validated source metadata, source confirmation and the current revision. Losing required permissions blocks further administrative actions.

Database Design

Firestore holds business documents and subcollections. IDs and participant arrays link records. Supabase migrations configure private storage policies rather than relational marketplace tables.

Table 3 Primary data collections

Collection or record group	Purpose
users and directory	Private account/personal profile and a limited name-search directory.
roles and verifications	Owner-managed privileges; identity state, submission timestamps and feedback.
listings and listingPrivate	Animal, seller, price, health declaration, media/status and protected pickup coordinates.
orders	Buyer/seller participant IDs, checkout snapshot, status and fulfillment details.
conversations with nested messages	Participant metadata; text, sender, timestamps and read-position state.
users/{uid}/notificationReads	Per-account event identifiers marking notifications as read.
voiceCalls with nested signals; callState	Participant call lifecycle, recipient-specific signaling and busy locks.
marketReferences and marketReferenceEdits	Dated price observations/history; private revision and editor audit records.
userRatings with nested raters; ratingSummaries	One editable rating per rater; aggregate count/sum for the target customer.
uploads and mediaCleanup	Media ownership, lifecycle tracking and cleanup/retry records.

Relationships and access

Orders link buyers/sellers to listing snapshots. Messages/calls use participant IDs; ratings use one record per rater. Private records restrict access, trusted changes use the backend and notification read state permits owner-only writes. Signed-in customers can view the profile's displayed contact/address fields.

Media handling

Public photos use Cloudinary; private IDs/vaccination files use Supabase. ID images are removed after review, withdrawal or 30-day expiry. Listing images have a maximum 1,600-pixel side and upload sequentially.

API Design

Trusted requests use HTTPS POST to **/functions/v1/animarket**, with a Firebase Bearer token and a JSON envelope containing **path**, **body** and logical **method**. The methods below describe envelope actions, not separate public HTTP routes. Firestore subscriptions handle authorized live reads; service credentials remain on the backend.

Table 4 Trusted API interfaces

Logical path or family	Action	Responsibility
/session/bootstrap	GET	Enable the Firebase-authenticated private-storage session role.
/auth/me; /auth/email; /auth/photo	PATCH /POST	Save authorized profile/email/photo changes; email requires recent password confirmation.
/verification/id; /verification/withdraw; /verification/reviews/{uid}	GET /POST	Submit/withdraw a pending ID; view private previews, the reviewer queue and decisions.
/uploads; /listings/publish; /listings/{id}/save, remove, pin	POST /GET	Validate upload ownership, publication, owned changes and protected pickup-pin access.
/orders/save	POST	Recheck identity, availability, totals, participants and allowed state changes.
/conversations; /conversations/{id}/messages, read	POST	Create/reuse private conversations, send deduplicated text and advance read state.
/calls; /calls/{id}/accept, decline, end, signals	POST	Authorize call lifecycle and private voice signaling.
/users/{uid}; /users/{uid}/rating	GET /POST	Return the permitted customer profile and save an editable 1-5 star rating.
/admin/market- references/{marketId}/prices/{priceId}	POST	Publish a source-checked price revision and preserve an audit record.

Development Process

Development is iterative and organized by feature. Routes connect screens to services, domain rules define valid behavior and data adapters implement persistence. Git and automated checks track incremental changes and guard architecture boundaries.

Git history begins on 22 September 2026, with authentication/marketplace work on 23-24 September. Cloud preparation and administration appear on 2 October; profiles/staff pricing on 3 October; address, identity and messaging changes on 4 October; logout/web fixes on 5 October. The reviewed revision is **9cbf427**, configured as version 1.0.0. Commits alone do not establish a completed public release.

CHAPTER 3

Testing

Testing strategy

Validation combines static analysis, architecture/domain checks, local backend tests and cloud-workflow simulations. Separate scripts cover emulator rules and live integrations. Local tests validate logic rather than native audio or current hosted-service availability.

Current verification results

On 7 October 2026, **`npx --no-install expo lint`** and **`npx --no-install tsc --noEmit`** passed without diagnostics. **`npm run check:app`** passed all **21 check groups**, including cloud-record logic, price revisions and public profiles/ratings.

Table 5 Key test cases and evidence

ID	Area	Expected result	Evidence
TC01	Account forms and access	Validate credentials/consent; prevent duplicate accounts and unsafe account switching.	Automated pass
TC02	Address and identity	Reject incomplete/out-of-area addresses and block unapproved buying/selling.	Automated pass
TC03	Listings and private media	Enforce seller/media ownership, valid pins and restricted document workflows.	Automated pass
TC04	Orders	Recheck totals/availability; apply allowed roles/states; deduplicate retries.	Automated pass
TC05	Messaging and signals	Restrict participants, preserve unread messages and recover failed listeners.	Automated pass
TC06	Staff price revisions	Require roles/source metadata; reject stale revisions and preserve the audit.	Automated pass
TC07	Public profiles and ratings	Limit exposed fields; prohibit self-rating and duplicate count increments.	Automated pass
TC08	Navigation and forms	Validate recovery paths, keyboard visibility, calendar input and web layout rules.	Automated pass
TC09	Native audio and final acceptance	Confirm actual microphone, speaker, reconnect and two-device workflows.	Pending device tests

Verification limits

These are local regression results. Earlier recorded emulator/live-cloud checks were not rerun. Native build/device acceptance remains outstanding. TURN is not configured; cross-network voice is unverified and background incoming-call push is absent.

Deployment

Configured deployment

The app configuration is version 1.0.0 with Android package **com.example.animarket_final**, portrait orientation and a light interface. EAS profiles select **EXPO_PUBLIC_BACKEND=firebase** and the public Firebase, Supabase and Cloudinary settings. Development builds use the native call modules; preview/apk profiles produce internal APKs, and production produces an Android App Bundle.

The current cloud workflow does not require the legacy local server. Private backend credentials belong in service secrets and excluded local files. Native directories are generated through Expo configuration and plugins. The staff browser portal is hosted separately at **animarket-87354-admin.web.app**.

Build and run procedure

1. With Node.js 24+ and npm, install locked dependencies using **npm ci**.
2. Supply the intended public cloud environment and run **npm run check:app**.
3. Use **npm start** for Expo Go-compatible features, or **npm run start:dev** with an installed development build for native voice.
4. Create the configured Android build with **npx eas-cli@latest build --platform android --profile apk**.
5. Install the signed build, complete device acceptance and use the production AAB profile for a later store release.

EAS configuration and successful JavaScript checks establish build readiness, not a completed APK or store publication. Repository build notes leave the native build and physical-device validation outstanding. Expo Go does not include this app's native WebRTC call modules.

User Acceptance

Acceptance should use two customer accounts and one authorized staff account on installed builds. Confirm registration and retained sign-in; address completion and ID approval; photo uploads and private documents; discovery and listing controls; pickup/delivery checkout; seller actions and buyer completion; real-time messages; notifications; public profiles/ratings; and staff price revisions.

Test denied and granted GPS/camera/microphone permissions, keyboard behavior, interrupted uploads, background/foreground changes, lost connectivity and sign-out. For voice, test two-way audio, mute, speaker, decline, hangup and separate networks. Accept a scenario only when the UI matches persisted records, prevents unauthorized actions and recovers without duplicate requests. Record device/build, scenario, expected result, actual result and pass/fail status. A completed UAT sign-off is still required.

Maintenance

Maintenance prioritizes accurate listings, authorized transactions and reliable cloud access. The backend includes ID expiry and orphan-media cleanup. Administrators should inspect cleanup failures and quota-related errors.

Table 6 Maintenance plan

Area or trigger	Required action
Defects and releases	Reproduce issues, fix the affected layer and rerun local/security regressions before release.
Access and media	Review staff permissions, Firestore/storage rules, service secrets and cleanup failures.
Market information	Publish identifiable live-weight sources with date, unit, revision and source confirmation.
Services and recovery	Monitor service errors/quotas; review retention, backups and restoration procedures.
Platform upgrades	Use matching Expo docs and SDK-compatible installs; rebuild and retest native features.
Documentation and support	Update changed workflows/schema/test evidence; collect reproducible user reports.

Documentation updates and user support

Update the version, geography, workflows, API/data model, test evidence and limits when behavior changes. User reports should include the build, device, screen, reproduction steps and error, and should be directed to the project team. A dedicated in-app helpdesk/reporting workflow is not implemented.

Conclusion

AniMarket version 1.0.0 provides a cloud-backed livestock marketplace for Davao del Norte. It connects discovery, approved-account buying and selling, private communication, dated price guidance and persistent order coordination. Server-side rules protect ownership, participants, ID documents and order transitions; current regression checks pass.

Future work

The next priorities are a completed native build and recorded device acceptance, production voice connectivity with TURN if required, background incoming-call notifications, and maintained price-source updates. Wider geography, structured offers, password-recovery UI and payment/logistics integrations are future features. Existing payment, transport, health and rating limitations should remain explicit until their behavior changes.

References

AniMarket Project Team. (2026). *AniMarket final project documentation*. Dwayne Rubite and Chester Emutan, Department of Computing Education, UM Tagum College. Supplied project reference PDF.

AniMarket Project Team. (2026). *Mobile_App_Final_AniMarket* (Version 1.0.0, source revision 9cbf427). Current source and configuration, reviewed 7 October 2026.

https://github.com/drockx/Mobile_App_Final_AniMarket

Implementation evidence: **package.json**, **app.json** and **eas.json**; **src/app**, **src/features** and **src/data/firebase_repositories.ts**; **supabase/functions/animarket**; **firestore.rules**; **supabase/migrations**; **scripts/check_app.js** and the scripts it invokes; **data/market_references**. Build and operational context: **docs/app_build_readiness.md**, **docs/backend_connection.md** and **docs/voice_calls.md**.

Expo. (2026). *Expo SDK 57 reference*.

<https://docs.expo.dev/versions/v57.0.0/>

Expo. (2026). *Introduction to Expo Router*.

<https://docs.expo.dev/router/introduction/>

Expo. (2026). *Introduction to development builds*.

<https://docs.expo.dev/develop/development-builds/introduction/>

Google Firebase. (n.d.). *Writing conditions for Cloud Firestore Security Rules*.

<https://firebase.google.com/docs/firestore/security/rules-conditions>

Supabase. (n.d.). *Firebase Auth*.

<https://supabase.com/docs/guides/auth/third-party/firebase-auth>

Version context

Version 1.0.0 and source revision 9cbf427 identify the application snapshot. Local regression results are dated 7 October 2026. Hosted data can change through authorized customer/admin actions. Market observations retain their own dates and source metadata; those dates must not be treated as current spot-price guarantees.